

Lab 9

Introduction to Predicate logic and Queries in Prolog

Example problem

Facts:

- 1) Cat is a mammal
- 2) Dog is a mammal
- 3) Eagle is a bird
- 4) Sparrow is a bird

Rules:

- 1) An Animal is a bird if it is a bird

is_a_bird(Animal) :- is_a(Animal, bird).

Methodology:

- 1) Create a file with extension **.pl** and write down the above facts and rules in that file
- 2) Save the file.
- 3) Open SWI-Prolog
- 4) In the **file** menu, select **consult**, open the **.pl** file you just saved above. If the facts and rules are loaded correctly then you will see the rules file loaded message
- 5) Now test the following using queries
 - a) Is eagle a bird?
 - b) Is cat a bird?
 - c) Which animals are birds?

Report format

Title: Title as above

Date: Lab assigned date

Objective

- To implement a predicate logic based system using Prolog

Theory

- Prolog programming language
- Predicate logic

Implementation

- Write down the implementation steps for solving the problem above
- Paste the screenshots of your outputs for provided queries

Conclusion